



The model ST-8000-12000-100 is a multi octave high power amplifier operating between 8000MHz and 12000MHz and offering a wide dynamic Range with 100 Watts min saturated power. The employment of gallium nitride (GaN) and chip-and-wire technology in manufacturing ensures this module state-of-the-art power performance with excellent power-to-volume ratio. It is ideal for jamming, EMC, test and measurement applications.

The module is equipped with alarm functions for over temperature and

Features:

- ◆ Small Size and light weight;
- ◆ Instantaneous ultra broadband;
- ◆ 50 Ohms Input and Output matched;
- ◆ Built-In control and protection circuits.

Electrical Specifications (+28VDC, 25° C, Pin=0dBm , 50Ω)

Parameter	Symbol	Min	Typ	Max	Units
Operating Frequency	BW	7800		12300	MHz
RF Output Power @PSAT	PSAT	100		160	Watt
Power Gain @PSAT	Gp	50			dB
Power Gain Flatness @PSAT	Δ Gp		±1		dB
Input Return Loss	S11			-10	dB
Harmonics @100W	H		-50		dBc
Spurious Signals @PSAT	Spur		-60		dBc
In/Output Impedance	Impedance		50		Ω
Operating Voltage	Vdc		28		Volt
Quiescent Current	IQ		8.1		Amp
Dynamic Current @50W	IDD		15.5		Amp
Power add efficiency	Eff	22	30		%

Absolute Maximum Rating

Parameter	Rating
Input RF drive level without damage	+10 dBm Maximum
Load VSWR @ POUT =100W	5:1 @ all load phase & amplitude continuous
Thermal Overload	85° C ±5° C shutdown Maximum

Environmental Specifications (Design to Meet)

Parameter	Minimum	Typical	Maximum	Units	Notes
Operating Temperature	-20		50	° C	
Non-operating Temperature	-25		65	° C	Storage
Relative Humidity (non-condensing)			95	%	

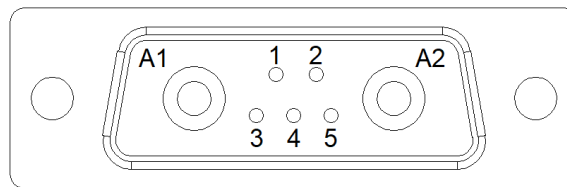
Mechanical Specifications

Parameter	Value	Units	Notes
Dimensions	160x120x25 [6.30x4.72x0.98]	mm [inch]	Maximum
Weight	2.0 [4.4]	kg [lbs]	Maximum
RF Connectors Input	SMA, Female		
RF Connectors Output	N-K, Female		
DC Interface Connector	Hybrid, 7W2 Mixed Contact D-Sub, Male		

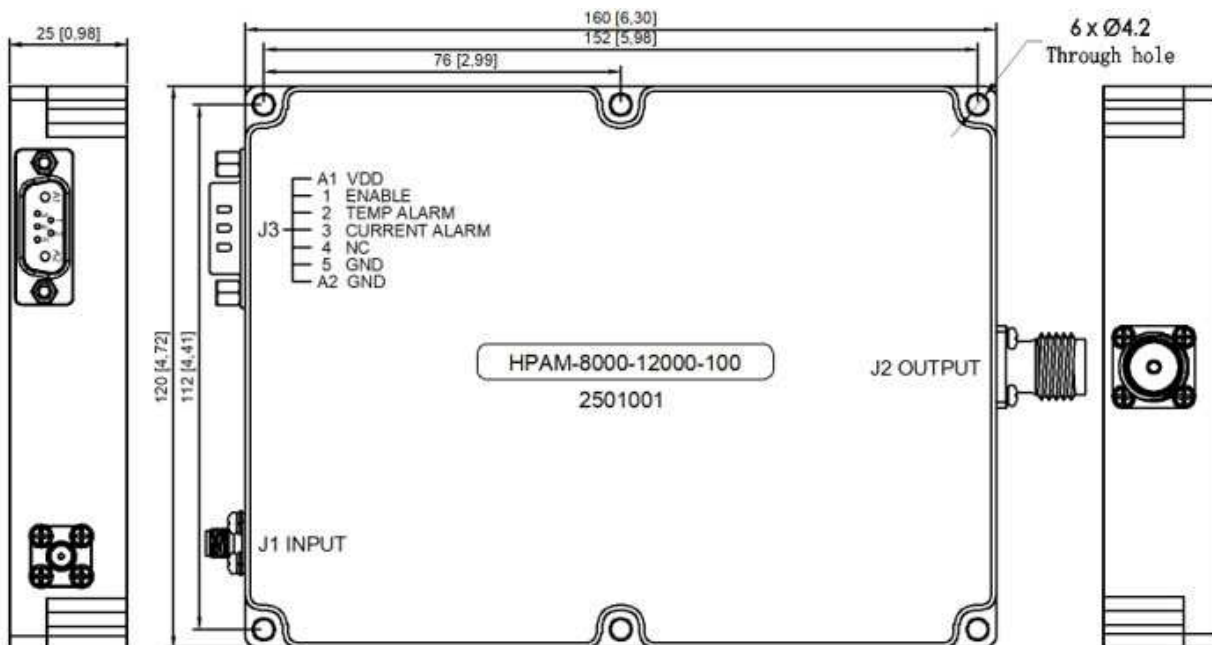


DC INTERFACE CONNECTOR

Pin #	Description	Specifications
A1	VDD	+28VDC
A2	GND	Ground
1	ENABLE	Amplifier Enable: TTL Logic High (3.3V) (Internally Pulled-Low) Temperature Protection, LVTTTL 0/+3.3V Output 0V: normal temperature
2	TEMP ALARM	+3.3V: the module temperature more than +85°C When the module temperature is higher than +85°C, the module shuts down automatically; After the temperature returns below +65°C, the module will turn on automatically
3	CURRENT ALARM	Current Protection, LVTTTL 0/+3.3V Output 0V: normal current +3.3V: over current When $I_{DD} > 20A$, 5s continued, the module shuts down automatically; the module will not turn on automatically. Users need to repower the module
4	NC	
5	GND	Ground



Outline Drawing (All dimensions in mm [inch])



Test Curve (Pin=0dBm, Room Temperature)

